

UNIVERSITY OF WESTMINSTER

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Module Title: Mathematics for Computing
Module Code: 4COSC002W.1
Exam Period: 16 January 2025
Time allowed: 120 minutes

INSTRUCTIONS FOR CANDIDATES

- Students who arrive more than 30 minutes late will not be permitted to enter the examination room and must apply for Mitigating Circumstances.
- Exiting the examination room is not allowed during the first 30 minutes and the last 15 minutes of the exam period.
- You are advised (but not required) to spend the first ten minutes of the examination reading the questions and planning how you will answer them.
- Students are not allowed to visit the toilet unaccompanied. If you need a comfort break during the examination, please be escorted by an invigilator. We ask everyone to minimise disturbances when moving in and out of rows.
- Calculators may be used provided they are silent, cordless, not pre-programmed by the candidate and cannot receive or transmit data remotely.
- It is mandatory to answer ALL QUESTIONS. The marks allocated for each task are indicated within the task's description.

The following information is given:

Formulae Sheet – Appendix 1

MOCK EXAM

**THIS PAPER MUST NOT BE TAKEN OUT OF THE
EXAMINATION ROOM**

**DO NOT TURN OVER THIS PAGE UNTIL THE INVIGILATOR
INSTRUCTS YOU TO DO SO**

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PART A – SHORT ANSWERS

(Provide concise, direct answers to each question. No additional explanations or workouts should be provided. Marks will be attributed based on providing the requested results, not the workouts.)

QUESTION A1 (5 MARKS)

Fill the gaps in the given statements:

- A. A matrix that has 1's on the diagonal and 0's elsewhere is called an _____ matrix.
- B. A matrix where all elements off the main diagonal are zero, but the diagonal itself can contain non-zero elements, is known as a _____ matrix.
- C. The probability of rolling a specific number on a six-sided die is an example of _____ probability.
- D. A graph is connected if, for every pair of nodes, there is a _____ between them.
- E. The depth of a node in a tree is the number of _____ from the root to that node.

QUESTION A2 (8 MARKS)

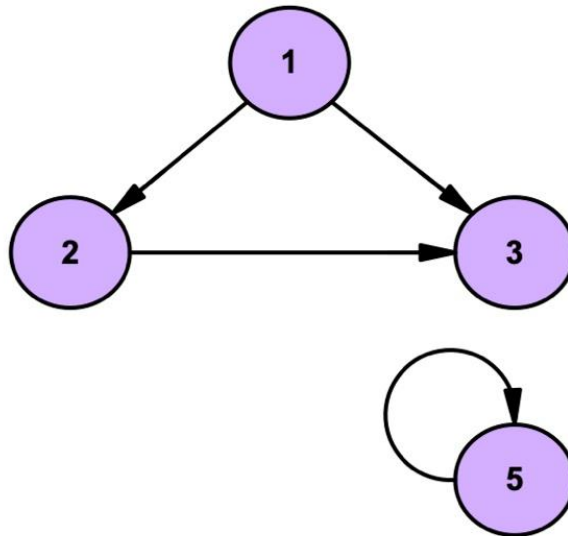


Figure 1 – Graph N

For the following **graph N** represented in *Fig. 1*, do the following:

- form *graph notation* for the given **graph N**: a set **V** of all its nodes and set **E** of all its edges;
- determine if it is a *directed* or *undirected* graph;
- determine if it is a *connected* or *disconnected* graph;
- determine if it is a *cyclic* or *acyclic* graph;
- determine the *degree of each node* and the *sum of the degrees of all nodes*.

QUESTION A3 (7 MARKS)

You are given the following equation, where X is an unknown 2×2 matrix:

$$3X - \begin{pmatrix} 2 & 5 \\ 3 & 6 \end{pmatrix} = 7X + \begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix}$$

Find the values of X_{11} and X_{22} in the matrix X that satisfy this equation. Assume that indexing starts at 1.

QUESTION A4 (8 MARKS)

Given matrices A and B :

$$A = \begin{pmatrix} 6 & -3 & 2 \\ 4 & 5 & -1 \end{pmatrix} \text{ and } B = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$$

Calculate the matrix C such that $C = A \times B$.

Provide only the elements C_{11} and C_{21} in your answer. Assume that indexing starts at 1.

QUESTION A5 (6 MARKS)

A software company has developed 100 programs with the following distribution:

- 25 programs written in Python.
- 30 programs written in Java.
- 20 programs written in C++.
- The remaining programs are written in JavaScript.

Find the PROBABILITY of selecting a program that is NOT written in Java.

QUESTION A6 (6 MARKS)

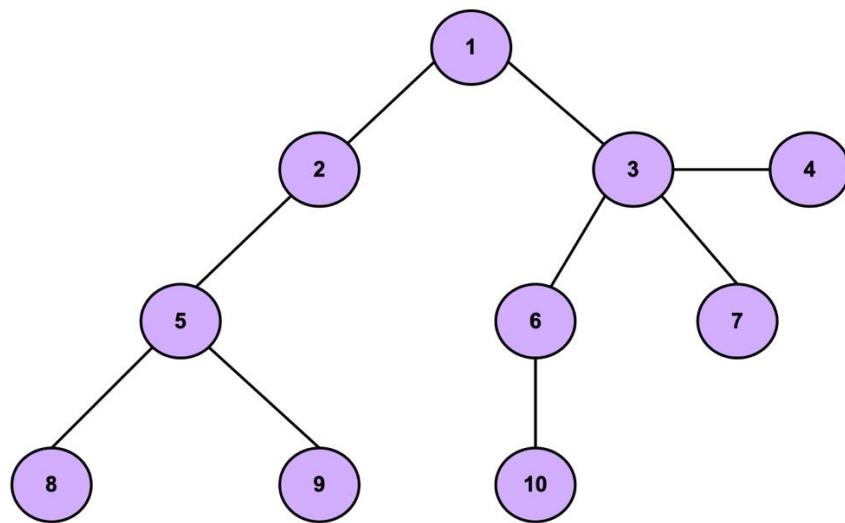


Figure 2 – Tree H

Tree H with the root node 1 is given in *Fig. 2*.

Identify:

- A. all the leaves of a tree;
- B. depth of a tree;
- C. is the given tree a binary tree (yes/no)?

PART B – DETAILED WORKOUTS

(Provide a comprehensive, step-by-step workout of each question. Each step should include relevant formulas, calculations, and brief explanations where necessary to justify your approach).

QUESTION B1 (15 MARKS)

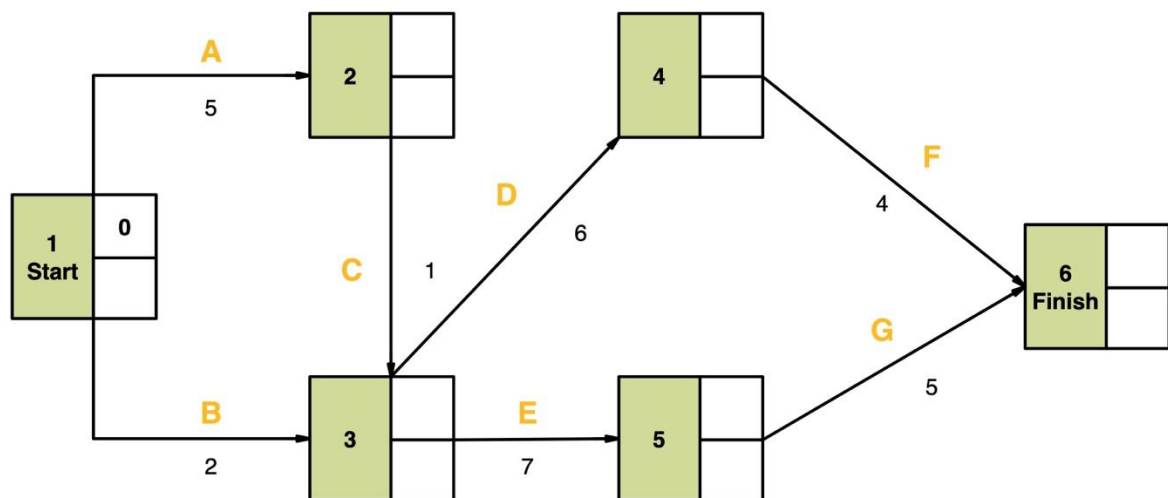


Figure 3 – Network Graph

The provided diagram in *Fig. 3* represents a project schedule using a network graph, where each enumerated node (or box) represents a *milestone* within the project. The edges represent *activity* and are labelled with letters. The duration of each task is given below the edge. The arrows indicate the sequence of tasks, demonstrating which tasks must be completed before others can begin.

Using the provided network graph, your objective is to perform a critical path analysis. This will involve identifying the longest path through the network and determining when the project can be completed.

For your analysis, follow these steps:

- A. Redraw the provided network graph (Fig. 3) on your answer booklet. Fill in all the empty cells (EST and LFT) for each node.

- B. Provide detailed calculations of the Earliest Start Time (EST, the upper empty cell of the node) and the Latest Finish Time (LFT, the lower empty cell of the node) for ALL tasks.
- C. Identify the critical path, which is the path from Start to Finish that has *zero slack* for all tasks. Show the critical path on your drawn graph. If there is more than one critical path, name at least one.

QUESTION B2 (13 MARKS)

A fair die is rolled twice, yielding two outcomes: X_1 (the result of the first roll) and X_2 (the result of the second roll). Consider two events:

Event A: $X_1 = 5$ or $X_2 = 5$ (at least one of the rolls results in a 5).

Event B: $X_1 + X_2 = 8$ (the sum of the two rolls is 8).

Find the CONDITIONAL PROBABILITY $P(A|B)$, which is the probability of *Event A* occurring, given that *Event B* has occurred.

QUESTION B3 (15 MARKS)

Analyse network latency by examining the time taken for packets to travel to a server and back. The dataset below records the round-trip times (RTT) for packets in milliseconds.

Table 1: Packet Round-Trip Times (in milliseconds)

Packet	RTT (ms)
P1	45
P2	60
P3	45
P4	75
P5	60
P6	60

Calculate the following statistical measures:

- A. Mean: Average RTT.
- B. Range: RTT range.
- C. Median: Central RTT.
- D. Mode: Most frequent RTT.
- E. Standard Deviation: RTT consistency.
- F. Variance: Degree of RTT spread.
- G. Interquartile Range (IQR): Spread of the middle 50% of RTTs.

QUESTION B4 (17 MARKS)

You are given a 3x3 key matrix K with some negative entries and a message vector M . The message is encrypted by multiplying K with M to produce an encrypted vector E . Your task is to calculate the encrypted message E , find the inverse of K , and decrypt E to recover the original message vector M .

Key Matrix K :

$$K = \begin{bmatrix} 3 & -2 & 4 \\ -1 & 5 & -3 \\ 2 & -6 & 1 \end{bmatrix}$$

Message Vector M :

$$M = \begin{bmatrix} 7 \\ -4 \\ 9 \end{bmatrix}$$

Questions:

- A. Calculate the encrypted message vector $E = K \times M$.
- B. Compute the determinant of K .
- C. Confirm if the matrix K is invertible and justify your answer.
- D. Calculate the matrix of minors for K .
- E. Use the matrix minors to form the cofactor matrix.
- F. Transpose the cofactor matrix to compute the adjoint matrix.
- G. Use the adjoint matrix and determinant to find the inverse matrix K^{-1} .
- H. Use K^{-1} to decrypt E and recover the original message vector M .

Note: Show all steps clearly, including intermediate calculations for minors, cofactors, and the adjoint matrix.

APPENDIX 1 – FORMULAE SHEET

Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Mean (Arithmetic Average)	$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$
Standard Deviation (for a sample)	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$
Variation (for a sample)	$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$
Quartiles' Positions	$Q_{\text{position}} = \frac{Q}{100} \times (n + 1)$
Interpolation	$Q = \text{Value at } k + f \times (\text{Value at } k + 1 - \text{Value at } k)$